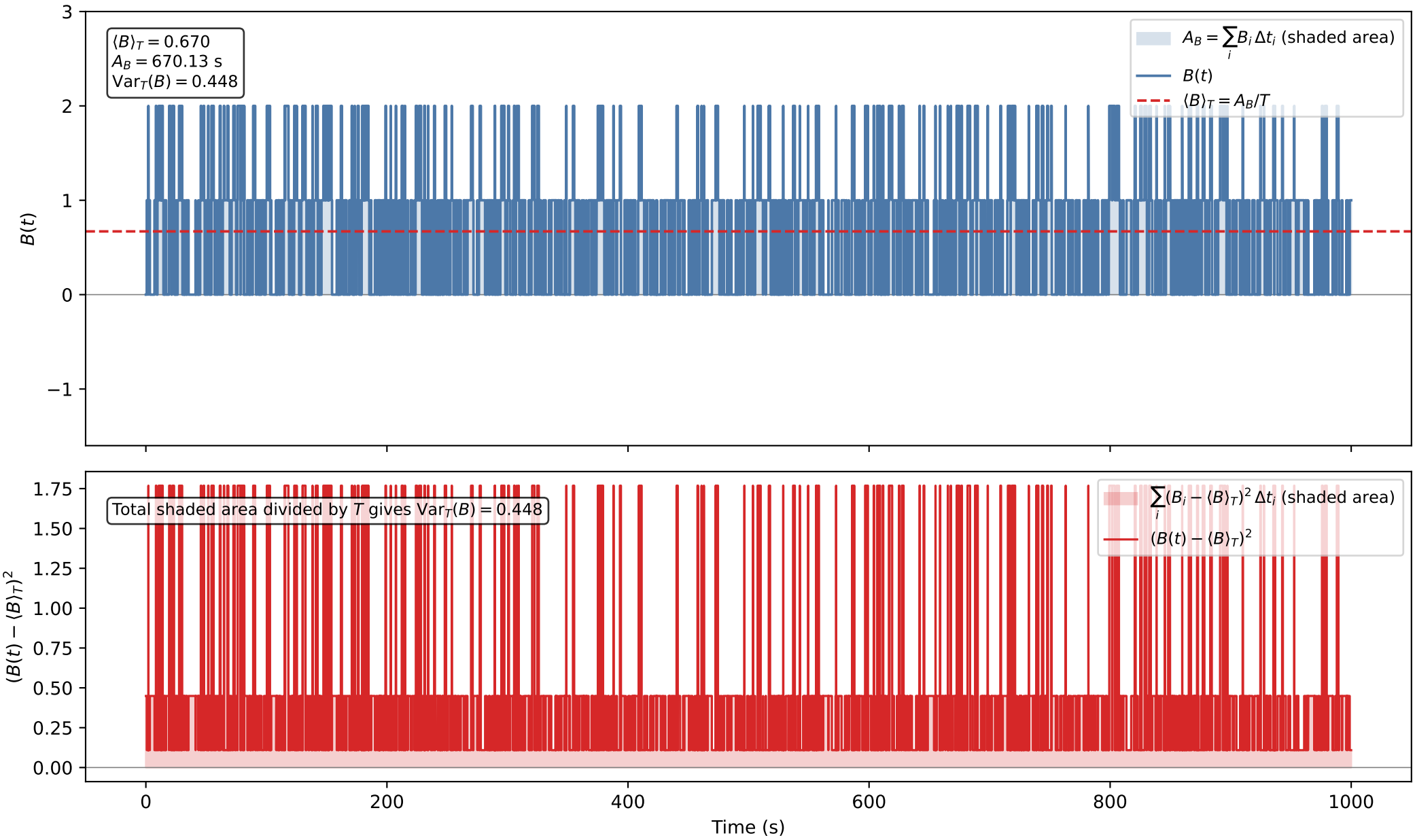

S2: construction of time-weighted mean and variance from one contact trajectory (r = 4, ag)
$\langle B\rangle_T = 0.670$
$A_B = 670.13$ s
$\mathrm{Var}_T(B) = 0.448$
$A_B = \sum_i B_i\,\Delta t_i$ (shaded area)
$B(t)$
$\langle B\rangle_T = A_B/T$
$B(t)$
Total shaded area divided by T gives $\mathrm{Var}_T(B) = 0.448$
$\sum_i (B_i - \langle B\rangle_T)^2\,\Delta t_i$ (shaded area)
$(B(t) - \langle B\rangle_T)^2$
$(B(t) - \langle B\rangle_T)^2$
Time (s)